



RAMCO INSTITUTE OF TECHNOLOGY

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Department of Computer Science and Engineering

Academic Year 2023 – 2024 (odd Semester)

Degree, Semester & Branch: B.E, V , CSE

Course Code & Title: CS3551& Distributed Computing

Name of the Faculty member (s): A.Sofia Thanga Mary

Innovative Practice Description

- **Unit / Topic:** Unit III / Deadlock detection in distributed system
- **Course Outcome:** CO3
- **Topic Learning Outcome:** TLO7
- **Activity Chosen:** One Minute Paper
- **Justification:**
Deadlock detection in distributed systems
- **Time Allotted for the Activity:** 15 Minutes
- **About the Event:**

A particular topic whose understanding is very important in the point of examinations and further understanding of the upcoming topics is chosen. This topic will be taught in the first 25 minutes of the class. At the end of the class, to analyse the level of understanding of the students, the students will be asked to write a few points regarding the topic taught in the class. From their write-up, the course instructor could know the level of understanding of students.

- **Details of the Implementation:**
 - Deadlock detection in distributed system was discussed in the class. Also, the System model of Deadlock detection were discussed.
 - At the end of the class, all the students were asked to write a few points about Deadlock Detection. The students wrote few points from deadlock detection up to their understanding level.
 - The content that they had written was verified at the end. After evaluating the one-minute paper written, the understanding level of the students were identified.
 - Around 10 % of students have written very few points from deadlock detection. A sample page written by haridharsana is given in Figure 1.
- **CO – PO / PSO**

CO	PO1	PO2	PO3	PO4	PO8	PO10	PO12	PSO1	PSO2	PSO3
	3	2	1	1	2	2	2	3	3	2

PO / PSO mapped:

Innovative practice	PO1	PO2	PO3	PO4	PO8
	3	2	1	1	2
Justification for correlation	The students could be able to get the knowledge in distributed deadlock detection using basic concepts of Computer Engineering.	Students will be able to identify, formulate and analyze the best solution for computer engineering problems with the basic knowledge of the distributed deadlock detection	Students could be able to provide the solution for complex problem with the avoidance of distributed deadlock.	Students will be able to use the knowledge of distributed deadlock detection in distributed system concepts, simple research problems could be solved in cloud computing using already existing algorithm	Students will be able to solve the mutual exclusion and distributed deadlock in distributed system.

PO10	PO12	PSO1	PSO2	PSO3
2	2	3	3	2
Students will be able to document the distributed mutual exclusion and distributed deadlock detection mechanism	Students could be able to continuously improve their knowledge in cloud computing concepts by completing some online course.	Students will be able to use the mutual exclusion, deadlock detection techniques, to avoid the common resource sharing problem when develop the software.	Students will be able to provide the distributed solution with the knowledge of mutual exclusion and distributed deadlock detection.	Students will be able to apply the distributed deadlock detection and mutual exclusion mechanism for analyzing the big data.

Images / Screenshot of the practice:

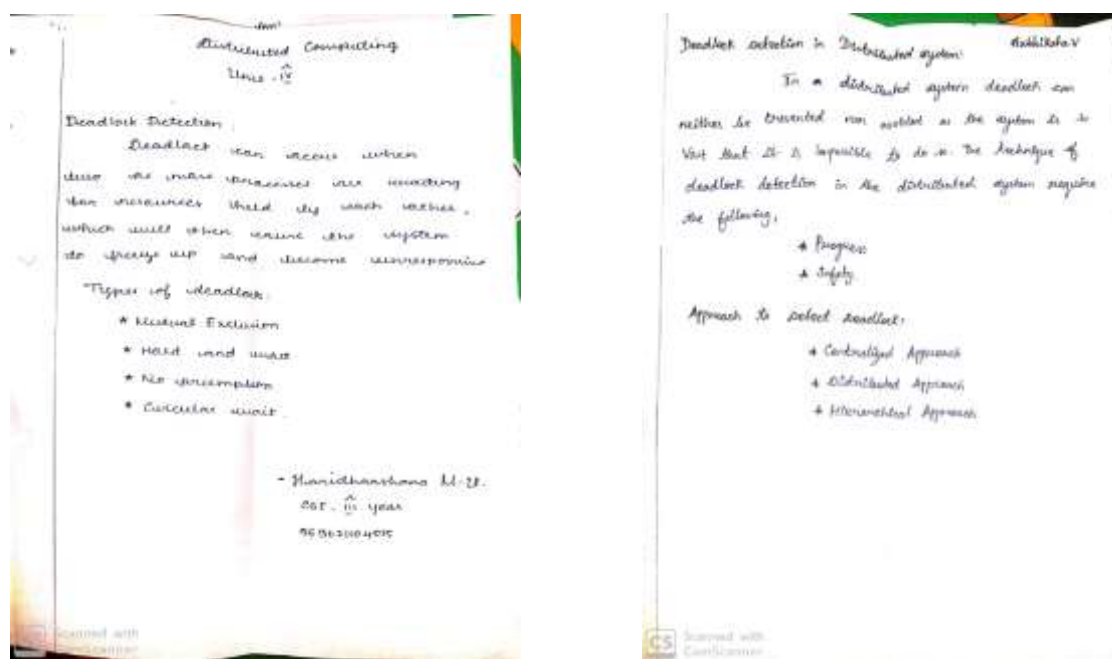


FIGURE 1. Sample Page written by students

Reflective Critique:

Feedback of practice from students and other stakeholders: The students gave good feedback about this activity. They felt that, they could be more thorough with this concept by doing this activity.

Benefit of the practice:

Most of the students take this activity as positive and started writing about Deadlock Detection in the paper.

- It can be easily tracked whether the students have understood the concept clearly or not.

Challenges faced in implementation:

- Very few students are not confident in their understanding towards this concept. They hesitated to prepare a write up on this concept.

Signature of Faculty Member

HOD